

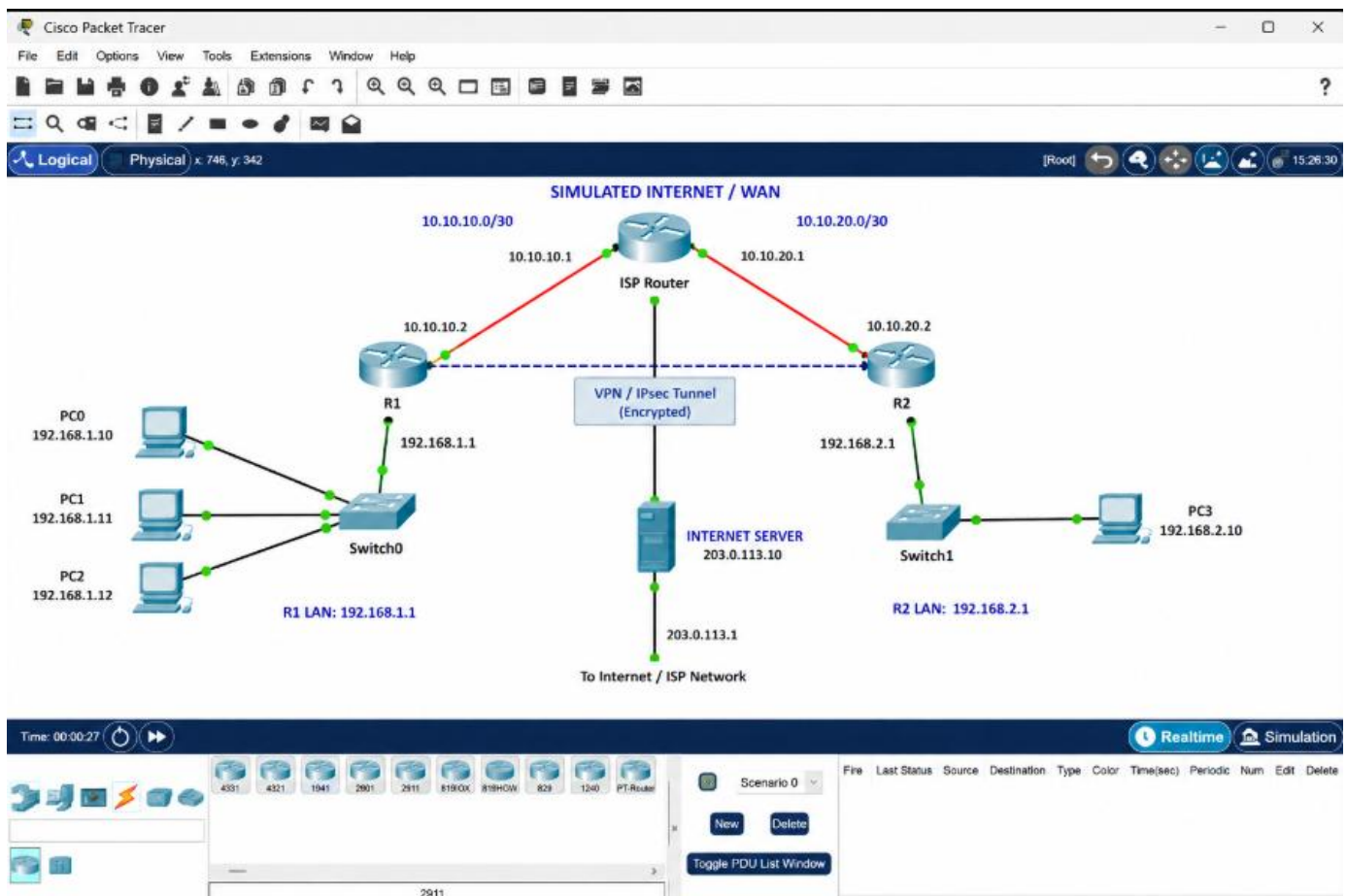
DAY_4

- ❖ **Weekly Practice and Troubleshooting Practical: Simulate a WAN setup with VPN and NAT.**
Task: Troubleshoot connectivity issues in the setup Checkpoint: Quiz: WAN technologies, VPNs, and NAT/PAT Practical Evaluation: Configure and troubleshoot a WAN.

1. **Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.**

ANS :

- Create a simulated WAN connecting **two branch offices** through a VPN tunnel. Configure **PAT/NAT** so internal devices can access an external Internet server. Then intentionally create VPN and routing problems and troubleshoot them using show and debug commands.



➤ **R1 Configuration**

```
enable
configure terminal
```

```
hostname R1
```

```
interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

```
interface gigabitEthernet0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
exit
```

```
ip route 0.0.0.0 0.0.0.0 10.10.10.1
```

➤ **R2 Configuration**

```
enable
configure terminal
```

```
hostname R2
```

```
interface gigabitEthernet0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

```
interface gigabitEthernet0/1
ip address 10.10.20.2 255.255.255.252
no shutdown
exit
```

```
ip route 0.0.0.0 0.0.0.0 10.10.20.1
```

2. Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address. Hint: Use the 'overload' keyword in your NAT configuration.

ANS :

Device	IP Address	Gateway
PC0	192.168.1.10	192.168.1.1
PC1	192.168.1.11	192.168.1.1
PC2	192.168.1.12	192.168.1.1

➤ **Configure NAT/PAT on R1**

```
enable
configure terminal
```

```
interface gigabitEthernet0/0
ip nat inside
exit
```

```
interface gigabitEthernet0/1
ip nat outside
exit
```

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

```
ip nat inside source list 1 interface gigabitEthernet0/1 overload
```

```
end
write memory
```

➤ **PAT Result**

```
PC0 192.168.1.10
PC1 192.168.1.11
PC2 192.168.1.12
```

— PAT Overload —→ 10.10.10.2

3. Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.

ANS :

➤ **Configure GRE + IPsec VPN**

- Step 1: Create the GRE Tunnel
- Step 2: Configure ISAKMP Phase 1
- Step 3: Configure IPsec

➤ **Router1**

```
enable
configure terminal
```

```
! =====
! STEP 1: GRE Tunnel Configuration
! =====
```

```
interface tunnel0
ip address 172.16.10.1 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.20.2
no shutdown
exit

! Route to Branch 2 through GRE Tunnel
ip route 192.168.2.0 255.255.255.0 172.16.10.2
```

```
! =====
! STEP 2: ISAKMP Phase 1
! =====
```

```
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
exit
```

```
! Pre-shared Key
crypto isakmp key VPNKEY123 address 10.10.20.2
```

```
! =====
```

HET

! STEP 3: IPsec Configuration

! =====

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
mode transport
exit
```

```
crypto ipsec profile VPN-PROFILE
set transform-set VPN-SET
exit
```

```
! Apply IPsec Protection to GRE Tunnel
interface tunnel0
tunnel protection ipsec profile VPN-PROFILE
exit
```

```
end
write memory
```

➤ Router2

```
enable
configure terminal
```

! =====

! STEP 1: GRE Tunnel Configuration

! =====

```
interface tunnel0
ip address 172.16.10.2 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.10.2
no shutdown
exit
```

```
! Route to Branch 1 through GRE Tunnel
ip route 192.168.1.0 255.255.255.0 172.16.10.1
```

! =====

! STEP 2: ISAKMP Phase 1

! =====

```
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
exit
```

```
! Pre-shared Key
crypto isakmp key VPNKEY123 address 10.10.10.2

! =====
! STEP 3: IPsec Configuration
! =====

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
mode transport
exit

crypto ipsec profile VPN-PROFILE
set transform-set VPN-SET
exit

! Apply IPsec Protection to GRE Tunnel
interface tunnel0
tunnel protection ipsec profile VPN-PROFILE
exit

end
write memory
```

4. Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

ANS :

➤ **Problem Scenario**

- Suppose PCs in Branch 1 can ping the Internet server:
PC0 → 203.0.113.10 = SUCCESS
- But cannot ping Branch 2:
PC0 → 192.168.2.10 = FAILED

➤ **Troubleshooting Procedure**

▪ **Step 1: Check Interfaces**

show ip interface brief

Verify:

- LAN interface is up/up
- WAN interface is up/up
- Tunnel0 is up/up

- **Step 2: Check Tunnel Status**

show interface tunnel0

Verify the tunnel source and destination are correct.

- **Step 3: Check VPN Status**

show crypto isakmp sa

show crypto ipsec sa

If the security association is not established, investigate the ISAKMP/IPsec configuration.

- **Step 4: Check Routing**

- On R1:

- show ip route

- The routing table must contain a route for:
192.168.2.0/24

- The correct route is:
S 192.168.2.0/24 [1/0] via 172.16.10.2

- If the route is missing, add:
configure terminal
ip route 192.168.2.0 255.255.255.0 172.16.10.2
end

- On R2, verify the return route:
show ip route
The correct route should be:
S 192.168.1.0/24 [1/0] via 172.16.10.1

- If missing:
configure terminal
ip route 192.168.1.0 255.255.255.0 172.16.10.1
end

➤ **Main Configuration Change That Resolves the Issue**

- The problem can occur when the branch-to-branch static routes are missing.

- **Add on R1**

- ip route 192.168.2.0 255.255.255.0 172.16.10.2

- Add on R2**

- ip route 192.168.1.0 255.255.255.0 172.16.10.1

- After adding both routes:

- Branch 1 PC → R1 → GRE/IPsec Tunnel → R2 → Branch 2 PC

- Connectivity should work.

5. Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it. Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.

ANS :

➤ **Example Misconfiguration Found**

```
show running-config
show ip route
show ip interface brief
show crypto isakmp sa
show crypto ipsec sa
```

- The missing route was:
ip route 192.168.1.0 255.255.255.0 172.16.10.1
- I added the route and tested the connection again using:
ping 192.168.1.10
- After adding the correct return route, the branch-to-branch connectivity was restored.

➤ **Complete Troubleshooting Commands**

```
show ip interface brief
show interfaces tunnel0
show ip route
show running-config
show running-config | section crypto
show running-config | include crypto isakmp key
show crypto isakmp sa
show crypto ipsec sa
show ip nat translations
show ip nat statistics
show access-lists
```